



DE Design Assurance

OE Overview



DESIGN ASSURANCE OVERVIEW

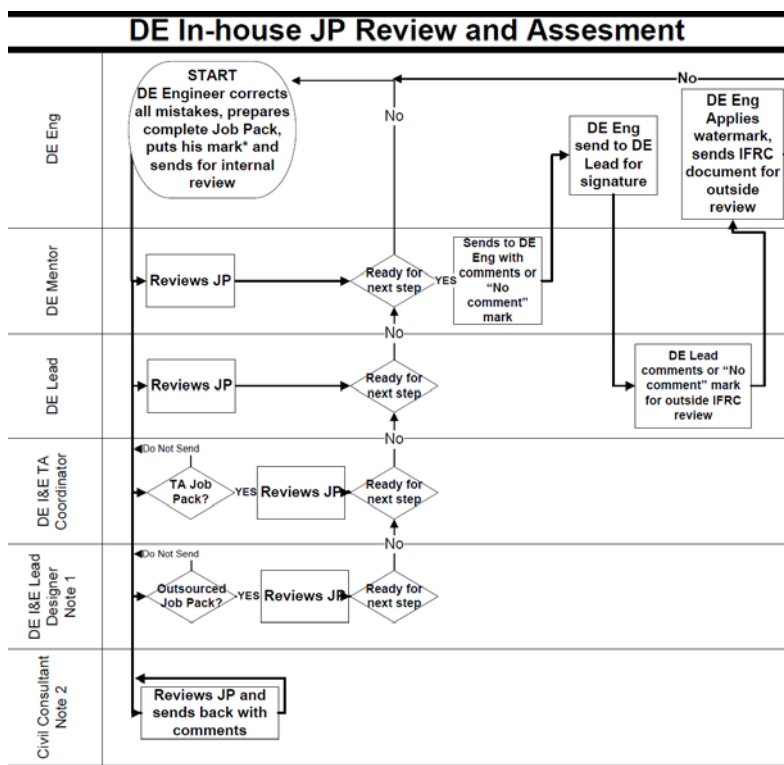
WHAT IS DESIGN ASSURANCE?

Design Assurance is the approach used to perform quality assurance / quality control of the engineering deliverables prepared by Designs Engineering (DE) and to capture relevant “Lessons Learned” from previous jobs, projects or incidents. Design Assurance will be performed on all DE Job Packs – it is a mandatory requirement to ensure appropriate technical reviews are completed prior to releasing the engineering deliverables for construction and installation.

JOB PACK REVIEW & APPROVAL PROCESS

All DE Job Packs will be subject to a Design Assurance review. The review process will utilize a discipline specific Design Engineering Design Assurance Checklist (DEDAC) and will be reviewed by the engineer during the development phase of the Job Pack (IFRC). DE’s internal Job Pack Review and IFRC Approval Process is shown below in Fig1.

FIG1.





DEDAC PROCEDURE

A step by step guide of the Job Pack review and approval process can be viewed in the following DEDAC OE guideline.

[DEDAC OE Guideline](#)

DEDAC – DESIGNS ENGINEERING DESIGN ASSURANCE CHECKLIST

The DEDAC is a discipline specific checklist. A Design Assurance tool for Engineers, Mentors and Leads to ensure adequate quality control is applied for all DE job packs. The appropriate DEDAC shall be selected based on the discipline type:

1. [Civil / Structural Checklist](#)
2. [Electrical Checklist](#)
3. [Instrumentation Checklist](#)
4. [Mechanical Checklist](#)

EXTERNAL REVIEWERS

All DE job packs are also reviewed and approved by external groups. Reliability Engineering, Maintenance, Process Automation, and Production Operations are included on the review cycle to confirm the engineering deliverables comply with the necessary requirements of these approver organizations.

DESIGN ASSURANCE ROLES & RESPONSIBILITIES

Design Engineer

- Design Engineer is responsible for utilizing the appropriate Designs Engineering Design Assurance Checklist (DEDAC) for the appropriate discipline.
- Design Engineers produce IFRC engineering Job Packs and complete applicable DEDAC's.
- Design Engineers are responsible for submitting the IFRC Job Packs firstly for internal review and then external review.
- Design Engineer is responsible for reviewing and incorporating all valid comments and recommendations on the IFRC Job Pack.
- Once IFRC stage is complete the engineer now follows the AFC revision process to achieve Approved for Construction status of the Job Pack.



DE Mentor

- The DE Mentor reviews the IFRC Job Pack for correctness to engineering deliverables and compliance with TCO Specifications and Standards.
- The DE Mentors Approval of a Job Pack indicates the engineering deliverables are correct and accurate. The DE Mentor approval also verifies that all requirements, and relevant Lessons Learned described in the DEDAC are included in the DE Job Pack. The act of signing the Job Pack by the DE Mentor confirms that design assurance of the engineering deliverables has been completed.

DE Lead Engineer

- Lead Engineer is responsible for final internal review of the IFRC Job Pack, and the associated DEDAC, the Lead Engineer either approves the job pack for external review or returns the Job Pack to the engineer for further revision.
- The act of signing the Job Pack by the DE Lead Engineer confirms that design assurance of the engineering deliverables has been completed.
- Lead Engineers are also responsible for auditing 2 Job Packs per rotation outside of their own work group.
- The Lead Engineer is responsible for communicating audit findings to their engineers and also addressing any high-risk findings immediately.

DE Advisor

- The Advisor is responsible for the facilitation of Job Pack Audit review meetings which shall be conducted once per rotation.
- Responsible for auditing 2 Job Packs per rotation.
- Reviews all Job Pack audit findings and facilitates discussion at Audit Results Review Meeting.
- Provides oversight and coaching regarding use of the DE Job Pack Auditing Templates. This includes identifying improvements that should be made to the work process or protocol to improve the effectiveness of this work process.

DE Supervisor

- DE Supervisor ensures Design Engineering Design Assurance process is being followed by attending DEDAC audit review meetings and assessing audit findings.
- The DE Supervisor is responsible for reviewing the efficiency and effectiveness of this work process on an annual basis.



AUDITING

Job Pack auditing is performed on a random sampling of DE Job Packs; the frequency and quantity of auditing on Job Packs will be adjusted on the analysis of the auditing results. Job Pack auditing is an additional measure of quality assurance / quality control on engineering deliverables prepared by Designs Engineering (DE).

DE Target is to audit approximately 10% of all DE AFC Job Packs completed in one year. The number of DE Job Pack audits completed by each position will be verified once per rotation during the Job Pack Audit Results Review meeting.

AUDITOR ROLE

The Auditor performs the job pack audit utilizing the appropriate audit template and documents the review results in the applicable audit folder. The Auditor shares the results of the audit with the applicable DE Lead Engineer who is responsible for the Job Pack. The auditor role is typically filled by a Lead Engineer, Mentor or Advisor.

1. [Electrical Auditors Checklist](#)
2. [Instrumentation Auditors Checklist](#)
3. [Mechanical Auditors Checklist](#)

AUDIT PROCEDURE

The Auditor Reviews Job Packs in accordance to the following OE Audit Guideline:

[Job Pack Auditing OE Guideline](#)

Job Pack selection criteria:

- a. Select from other DE group (Lead Engineer does not audit job packs his position has approved).
- b. Select Discipline Job Pack that Lead Engineer has relevant discipline responsibility.
- c. Issued Approved for Construction (AFC).
- d. AFC issued within last 12 months.
- e. Job Pack has not previously been audited.
- f. Not necessary that Job Pack has had construction completed status.

Job Pack Audit Results Review Meeting will be conducted once per rotation to review audit results and determine if any changes or improvements are required to Job Pack audit process or audit protocol. Significant audit findings will be dealt with immediately and shared with the Responsible Lead Engineer, Engineer and Mentor.



CONTINUOUS IMPROVEMENT

DE Advisor positions are responsible for improving the effectiveness and efficiency of this document and its associated work processes.